Statistical Research: Data Sources

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What one observation represents

The scoring unit is an NHL **player and regular season**. To assemble each row, we link five-on-five NHL events and shifts with the same players and games in the All Three Zones transition workbook. Recorded behavior covers 2021–22 through 2024–25, and roster outcomes extend through 2025–26 for a one-year follow-up.

NHL play-by-play supplies contact and shooting events, rosters identify players, and shifts measure five-on-five exposure; Corey Sznajder's tracked records add forechecking, retrieval, and entry-defense actions. We obtain NHL records through the pinned `nhlscraper` revision and use the All Three Zones glossary for the tracked measures. Retrieved on 2026-09-15, the workbook has SHA-256 fingerprint 0ed9c09819854e6f4b281730ba2d0b3e2dcc629a48ce6fc6fbf9bc35331907fd.

Official NHL draft-year scouting passages describe physical traits before the included NHL seasons. Public files retain coded mentions and a source catalog; passage text and blinded identity keys stay outside Git.

Matching and eligibility

We preserve the workbook's original labels while identifying games from teams, dates within three days, and NHL rosters. Names and sweater numbers help distinguish players who share a name. A game must match unambiguously and identify at least 90% of its players; we then retain one valid row per NHL player-game, collapse identical duplicates, and exclude conflicts.

Of the workbook's 1,699 game labels, 1,684 resolve to 1,683 distinct NHL games; fifteen labels remain unresolved. The player-game audit then yields the following flow.

Table 1: Disposition of All Three Zones skater-game rows after identity, count, and exposure checks.

Status	Rows
Retained	60,331
Unresolved game	540
Unresolved player	1
Invalid event counts	6
Player exposure disagreement	40
Game exposure disagreement	180

We next compare source ice time with NHL five-on-five exposure reconstructed from shift intervals. We exclude a player-game when its absolute exposure difference exceeds 120 seconds or its game's median absolute difference exceeds 60 seconds. Event counts must also be nonnegative integers, with entry denials no greater than targets. These checks retain 60,331 player-game rows, with a median absolute exposure disagreement of 11.00 seconds. We calculate rates using **NHL shift-derived minutes** and retain source minutes for audit.

At the player-season level, eligibility for scoring requires at least 300 full-season NHL minutes, listed height and weight, and a known birth date. Scoring also requires at least 150 matched five-on-five minutes, while published rankings will require 500 full-season minutes. Of 2,731 candidate player-seasons, 2,314 meet the tracked-exposure threshold: 1,485 forwards and 829 defensemen. The remaining 417 lack enough tracked exposure.

Table 2: Eligible scored player-seasons by position and behavior season.

Season	Forwards	Defensemen	Total
2021–22	399	215	614
2022–23	374	203	577
2023–24	359	204	563
2024–25	353	207	560

The included player-seasons represent 825 players. Their median matched exposure is 280.92 minutes for forwards and 351.18 minutes for defensemen. We model each season separately.

Variables used to construct CSAx

Both positional models use the same six direct measures. For each, we divide a player's attributed NHL events in matched five-on-five games by that player's NHL five-on-five minutes and express the rate per 60. With positive exposure, a recorded zero remains zero.

Measure	Recorded numerator and interpretation
Hits delivered	Hits attributed to the player as hitter; possession and role affect opportunities.
Hits received	Hits attributed to the player as recipient; exposure to contact need not mean initiating it.
Opponent shots blocked	Shots blocked for the defending team, excluding teammate blocks; a recorded block need not involve body contact.
Fights	Recorded fighting infractions, counted separately from contact penalties; events are sparse.
Contact penalties taken	Whitelisted contact-related infractions attributed to the player, counted as incidents rather than minutes.
Contact penalties drawn	Whitelisted infractions committed by opponents against the player; puck possession and skill also affect exposure.

The contact-penalty whitelist covers boarding, charging, checking from behind, clipping, elbowing, head checks, kneeing, roughing, slew-footing, cross-checking, high-sticking, holding, holding the stick, hooking, and tripping, with recorded helmet-removal and double-minor variants. Fighting is counted separately. The exact source labels are fixed in `R/functions.R`.

Position-specific measures add information about involvement in contested space, where forwards and defensemen face different opportunities. They do not record the pressure a player faces at every moment.

Forward measure	Unit and denominator	Interpretation
Net-front attempt share	Located unblocked attempts at $82 \leq x \leq 89$ and $\lvert y \rvert \leq 8$ feet / all located unblocked attempts	Shooting near the crease; location does not confirm pressure.
Tip/deflection share	Tips and deflections / shots on goal or goals with known shot type	Redirections often occur near traffic; role affects opportunities.
Median shot distance	Feet from goal among located unblocked attempts	Proximity of shooting involvement.
Backhand share	Backhands / shots on goal or goals with known shot type	Possible constrained shooting space; backhands occur elsewhere.
Dump-in recoveries	All Three Zones `Recoveries` per 60 matched minutes	Recovering a dump-in and making a play.
Forecheck pressures	All Three Zones `Forecheck Pressures` per 60 matched minutes	Pressuring a puck-carrying defender.

Defenseman measure	Unit and denominator	Interpretation
Defensive-zone retrievals	All Three Zones `DZ Retrievals` per 60 matched minutes	Collecting the puck in the defensive zone and making a play.
Botched retrievals	All Three Zones `Botched Retrievals` per 60 matched minutes	Retrieval-related breakdowns; the tracked category can involve the receiving player.
Entry denial share	All Three Zones `Denials` divided by `Targets`	Preventing an opponent's attempted entry; gap control and team tactics also matter.

The feature table retains source counts and share denominators. A share with no opportunities remains missing until training-sample imputation. The forward model does not use the undefined entry-denial share for one center-season, so all modeled primary features are present. A 20-opportunity threshold flags sparse denominators but does not guarantee reliability.

Outcomes and interpretation

Scouting codes record earlier mentions of active physical traits such as checking and puck battles. Silence cannot establish soft play. Official NHL records provide power-play and penalty-kill shares of total regular-season ice time; recorded zeros remain distinct from missing values.

For the postseason comparison, we will count five-on-five hits delivered and received and opponent shots blocked in each team's first four playoff games. Regular-season games excluded from CSAx inputs supply the baseline; rate comparisons require positive exposure in both periods. Continuation requires at least 300 NHL minutes the next season, with minutes per game measured among returners. We will adjust for listed size, age, current games and ice time per game, five-on-five points per 60, relative shot attempts, and season.

Tracked games are selected unevenly, and player-game totals lack timestamps or linked possession sequences, so retrievals and denials remain proxies for physical involvement. We interpret them alongside scouting evidence, roles, and coverage.

Reproduction and sources

Inputs and provenance are bundled in `data/analysis_data.rds`. Restore `renv.lock` and render this QMD with Quarto; fresh acquisition requires the All Three Zones workbook and pinned `nhlscraper`.